long-term data, which is not available easily. Dr Dastidar is hopeful though, and believes that with more and more awareness of the capabilities of AI in disease prediction, such data will be more abundantly available. "Developing AI for diagnostics is a very cross-functional effort, requiring equal participation from engineers, doctors and medical institutions," he says. Then, there is the issue of the quality of the diagnostics data of the individual being tested. All analytics ultimately depend on blood markers, body scans and genetic markers to make a prediction. If these are not of high quality, then the prediction will not be reliable. There is also the risk of false negatives.

Privacy of data is another hurdle. Several AI companies have access to a large volume of patient data and experts say it is extremely crucial that they implement strict security to prevent leakage of any kind. Right now, there are no strict regulatory or governance standards to manage what AI should or shouldn't do, or how data should or shouldn't be used.

"Developing AI for diagnostics is a **cross-functional effort, requiring equal participation** from engineers, doctors and medical institutes... large-scale clinical studies are needed"

–DR TATHAGATO RAI DASTIDAR
Founder & CEO, SigTuple



THE WAY AHEAD

Most doctors warn that predictive health tools provide only risk analysis, not a guarantee against getting the disease in the future. Once they ascertain the risk, steps can be taken to mitigate it. For example, a smoker who has a risk of developing lung cancer or COPD can opt for tobacco cessation. It may not fully eliminate the risk of disease but certainly delay its likely onset by 10-15 years. Likewise, someone with a high risk of cancer can go for more frequent screenings to catch it at an early stage rather than when it has progressed. Some medical interventions can also completely reverse lifestyle diseases. For example, in familial hyperlipidemia, cholesterol is always abnormally high, something genetic testing can determine. If the individual is put on statins on time, he or she can be spared a heart attack. "In some cases, if a person lives with utmost discipline and strict lifestyle rules, they can avert a disease completely without medical intervention. But this is quite rare," says Dr Shalini Joshi, an internal medicine specialist at Fortis Hospital, Bannerghatta Road, Bengaluru.

The real focus of predictive analytics in healthcare, therefore, is to set an individual on the path of course correction well in time to prevent—if not the disease—at least the seriously acute problems, irreversible damage and other complications that can arise from treatment. Interestingly, some doctors say that early risk prediction can rule out the need for medication altogether. "The irony is that some patients actually prefer medicine because it is a shortcut solution and you can continue to live the life you are living with little regard to your health," says Dr Randeep Guleria, the former director of AIIMS, Delhi, who is currently the chairman of the Institute of Internal Medicine and Respiratory and Sleep Medicine at Medanta. "But I don't think that is the right step. All drugs have side-effects. It is best to look at early corrective measures that are non-pharmacological to begin with and more lifestyle-oriented." These lifestyle modifications can include diet restrictions, yoga, exercise, meditation, regular health screenings, doctor follow-ups and so on. So, if you have a high risk of diabetes, you don't start with insulin, you first modify your diet and start exercising.

That is what 21-year-old Sudipto Chaudhuri, a management intern from Kolkata, has started doing. Genetic and blood tests as well as family history confirmed that he was a candidate for future heart attacks. Since he now knows this at an early age, doctors have advised him against statins—a group of medicines often prescribed to lower cholesterol—or heart medication. Instead, he is modifying his diet to reduce fat, and has started daily cardio exercises. He has also stopped drinking and smoking. "I was motivated to quit, knowing that if I continue, I will definitely die early. Knowing the risk has helped me," he says.

Doctors say predicting one condition can also help prevent other problems. So, managing one's blood pressure so that it does not become chronic can prevent heart attacks and strokes. In fact, a recent study at Medanta shows that childhood obesity can result in sleep apnea and other metabolic disorders in adulthood. "It is best to start investing in and be conscious of your health as early as possible," says Dr Guleria. While one cannot go back in time to prevent the onset of disease, thanks to predictive diagnostics and intelligent risk analysis, one can predict the future and take steps before it's too late. ■